

IRFAN ALI

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TECHNICAL SKILLS

Cloud (AWS): VPC, EC2, ALB, Auto Scaling, S3, IAM, RDS, Lambda, DynamoDB

IaC & CI/CD: Terraform, GitHub Actions, ArgoCD

Containers & Orchestration: Docker, Kubernetes, Helm

Policy & Security: Open Policy Agent, Kyverno

Real-Time Networking: WebRTC, RTCDataChannel, ICE/STUN/TURN, coturn, WebSockets, Node.js

Languages: Python, Go, Bash, YAML, HCL

PROJECTS

FizzShare — WebRTC Peer-to-Peer File Sharing github.com/irfanjat/fizzshare

TypeScript, WebRTC, RTCDataChannel, ICE, coturn, Node.js, Docker, Terraform, Prometheus

- Built a browser-to-browser encrypted file-sharing app in TypeScript over WebRTC RTCDataChannel (SCTP/DTLS across ICE), so file bytes travel directly between peers while the Node/Express server only relays signaling and mints short-lived TURN credentials.
- Shipped it live on AWS using Docker Compose (signaling, coturn, Caddy with automatic TLS) provisioned by Terraform, with a WebRTC inspector for ICE candidate pairs, RTT and packet loss, Prometheus metrics, and SHA-256 integrity verification on every transfer.

End-to-End GitOps CI/CD Pipeline github.com/irfanjat/gitops-cicd-pipeline

GitHub Actions, ArgoCD, Kubernetes, Docker, Helm, Python

- Designed a GitOps deployment pipeline with GitHub Actions and ArgoCD that replaced manual kubectl deployments with push-to-deploy releases using SHA-tagged, immutable Docker builds and Helm charts.
- Configured ArgoCD for automated sync, drift detection, and self-healing, enabling continuous reconciliation and rolling updates across the Kubernetes cluster.

DB Operator — Kubernetes Operator for PostgreSQL github.com/irfanjat/db-operator

Go, controller-runtime, Kubernetes CRDs, Helm, ArgoCD, Prometheus

- Built a custom Kubernetes operator in Go using controller-runtime that manages PostgreSQL clusters via Custom Resource Definitions, extending the Kubernetes API rather than relying on existing controllers.
- Implemented automated S3 backups, Prometheus monitoring, a Helm chart, and ArgoCD GitOps deployment for continuous reconciliation.

Policy Guardrails Engine github.com/irfanjat/Guardrails

Open Policy Agent, Conftest, Kyverno, GitHub Actions

- Built a policy-as-code guardrail engine enforcing security and compliance rules on Terraform plans (via Conftest/OPA) and Kubernetes manifests (via Kyverno) inside a CI pipeline.
- Integrated automatic pull-request comments on policy violations, blocking non-compliant changes before merge.

Production Multi-Tier AWS Infrastructure github.com/irfanjat/terraform-aws-infra

Terraform, VPC, EC2, ALB, Auto Scaling, RDS, S3, DynamoDB

- Provisioned a multi-availability-zone AWS environment of 24 resources using modular Terraform, including VPC, subnets, Application Load Balancer, EC2 Auto Scaling, and RDS with automated failover.
- Secured infrastructure state and access by establishing S3 remote state with DynamoDB locking and scoped, least-privilege IAM policies.

EDUCATION

University of Sindh (SULC) Jamshoro, Pakistan

Bachelor of Science in Computer Science · Expected 2027

CERTIFICATIONS

- OCI Foundations Associate — Oracle Cloud Infrastructure
- DevOps Engineering Professional Certificate — IBM